

GEOFFREY MAINA WACERA
TRANSACTION MONITORING ANALYST | MACHINE LEARNING ENGINEER

Nairobi, Kenya • mainageoffrey06@gmail.com • +254 703 110 996 • [GitHub](#) • [LinkedIn](#)

Profile

I am a solution-focused professional with hands-on experience in real-time monitoring systems, and automation engineering within a fast-paced fintech and large-scale financial institutions. Using SQL, PostgreSQL, Python, Grafana, low-code tools, and AI-driven RAG workflows, I work to keep systems efficient, accurate, and reliable. I enjoy building smarter automation processes that eliminate repetitive and manual tasks, improve the speed and accuracy, and ensure systems run seamlessly.

My earlier background in software development and frontline banking gives me a unique ability to understand challenges from both a technical and operational perspective. I've built Android apps, designed REST APIs, supported customers directly, and ensured compliance with strict AML/KYC standards—experiences that taught me the importance of accuracy, clarity, teamwork, and trust. I bring this blend of technical skill and human understanding into every project, aiming to create solutions that are not just efficient, but genuinely helpful to the people and processes they support.

Education

University of Nairobi Bachelor of Economics Second Class Honors (Upper Division)	Nairobi, Kenya December 2021
---	---------------------------------

Nairobi School KCSE Score: A- (80 points)	Nairobi, Kenya November 2015
---	---------------------------------

Experience

EQUITYBANK (K) LTD Transaction Monitoring Analyst	Nairobi, Kenya August 2024 – Present
--	---

- Conduct real-time evaluation of financial transactions using SQL, PostgreSQL, and Grafana to identify anomalies, operational weaknesses, and potential fraud indicators.
- Contribute to the development and refinement of the enterprise-wide transaction monitoring system, ensuring full and accurate transaction capture across core banking, mobile banking, agency, and card channels for all Group subsidiaries.
- Assess the effectiveness of automated monitoring rules, system logic, thresholds, and alert behavior to improve detection accuracy and reduce false positives.
- Leverage Grafana dashboards to monitor live transaction flows, activity thresholds, and anomalies, supporting data-driven decision-making.
- Perform ETL, SQL, and PostgreSQL investigations using joins, CTEs, and data extraction techniques to validate alerts, analyze trends, and determine root causes of anomalies.
- Build and deploy automation workflows using Python, SQL, low-code/no-code tools (n8n, Power apps and Zapier) and AI-driven RAG (Retrieval-Augmented Generation) agents to streamline processes and eliminate repetitive manual tasks.
- Develop automated scripts and intelligent workflows to extract, transform, and analyze transactional data, enabling faster anomaly detection and operational risk assessment.
- Collaborate with business units across all subsidiaries (Kenya, Uganda, Tanzania, South Sudan, and Rwanda) and compliance teams to escalate verified suspicious cases with clear evidence and analytical justification.
- Deliver immediate email alerts on high-risk transactions, reducing exposure to financial loss and strengthening the bank's fraud-prevention framework.
- Guide business units in designing report formats, advising on optimal data transformation and automation approaches, and developing reports and dashboards tailored to user specifications for improved decision-making and control.
- Monitor system uptime and proactively coordinate resolution of downtimes or processing delays to maintain continuous surveillance and operational integrity.
- Identify, analyze, and interpret trends or patterns in transactional data and prepare actionable reports.

GEOFFREY MAINA WACERA
TRANSACTION MONITORING ANALYST | MACHINE LEARNING ENGINEER

E&M TECHNOLOGY HOUSE LTD

Nairobi, Kenya

Software Developer

January 2024 – July 2024

- Led a diverse team of developers in designing and developing an automation solution project from inception to completion, ensuring successful delivery by collaborating across all phases of the software development lifecycle, from requirements gathering to deployment, to ensure a comprehensive understanding and execution of project goals.
- Coordinated and contributed to the development and maintenance of high-performance Android applications using Kotlin (Jetpack Compose and XML), utilizing MVVM architecture, Room database, Live Data, and other modern Android frameworks to enhance user satisfaction and scalability.
- Engaged with my team in agile development cycles by participating in daily stand-ups, sprint planning, and retrospectives to ensure timely project delivery and continuous improvement.
- Designed and implemented RESTful APIs with Java Spring Boot to support mobile app functionality and data exchange, thereby improving application efficiency.
- Delivered project progress presentations for internal review and to stakeholders, ensuring transparency and alignment with project objectives.

EQUITY BANK (K) LTD

Nairobi, Kenya

Relationship Officer

May 2021 – December 2023

- Delivered accurate domestic and international wire payments with full adherence to KYC, AML, and CTF regulatory requirements under the POCAMLA.
- Achieved a 100% compliance rating on cheque processing and liquidations, maintaining zero violations under Cheque Truncation guidelines.
- Efficiently delivered cash payments and receipts with precision, meeting 95% daily transaction targets on average and maintaining a 98% Service Level Agreement based on client feedback.
- Developed and maintained strong client relationships, decreasing client account dormancy rate from 30% to 13% through proactive engagements.
- Promoted digital products, increasing digitized clients by 22% and reducing branch traffic through active engagements.
- Provided ongoing training and development to team members on various bank platforms, enhancing team capabilities and performance.

Leadership and Activities

E&M TECHNOLOGY HOUSE LTD

Nairobi, Kenya

Project Lead

January 2024 – July 2024

- Led a diverse team of developers to design and develop a solution project from inception to completion, ensuring successful delivery and collaboration.

Skills and Interests

Technical Skills:

- Cloud and ML: Certified AWS ML Engineer Associate, model workflows, cloud pipelines
- Data analysis and Automation: Python, SQL and PostgreSQL ETL, low-code automation (n8n), AI-powered RAG agents
- Software Engineering: Kotlin (Jetpack Compose, XML), Java Spring Boot, mobile and backend systems
- Integration and System Design: REST APIs, MVVM, Room Database
- Monitoring and Dashboards: Grafana, Excel, data visualization

Soft Skills:

- Analytical Thinking and Problem Solving
- Leadership and Team Collaboration
- Attention to Detail and Accuracy
- Communication and Stakeholder Management
- Adaptability and Learning Agility

Languages:

- English (Fluent)
- Swahili (Fluent)
- French (Basic)